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Exercise 2: SQL Foundations
SQL Aggregate Functions & Operators

① The students Table

1. List all distinct departments in the student table

```
SELECT DISTINCT departments
FROM students;
```

department
IT
HR
Finance

2. Get the average age of students per department

```
SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
Finance	23

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3. Show departments with more than 1 student

```
SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) >1;
```

department	student_count
IT	2
HR	2

4. Get all students whose age is between 21 and 23

```
SELECT *
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. List all students in the IT or HR department who are older than 21

```
SELECT *
FROM students
WHERE department IN ('IT', 'HR')
```

③

AND age > 21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

② The Courses Table

6. Show total credits per departments, only for departments with more than 5 total credits

```
SELECT department,
       SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING total_credits > 5;
```

department	total_credits
IT	11

7. List all courses that do not have 4 credits

```
SELECT *
FROM courses
WHERE credits <> 4;
```

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course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Show the top 3 courses by credits descending order

```
SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

3 The enrollments Table

9. Get the maximum, minimum and average grade across all enrollments

```
SELECT MAX (grade) AS max_grade,
       MIN (grade) AS min_grade,
       AVG (grade) AS avg_grade
FROM enrollments;
```

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max_grade	min_grade	avg_grade
90	78	84.6

10. Count how many enrollments exist per course

```
SELECT course_id, COUNT(enrollment_id)
        AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

4 The salaries Table

11. Find total salary and total bonus per department

```
SELECT department,
        SUM(salary) AS total_salary,
        SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

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department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. Show departments where average salary is above 55000

```
SELECT    salary,
          AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000
Finance	70000

13. List employees whose salary plus bonus is greater than 60000

```
SELECT    employee_id, nam, salary, bonus,
          (salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000;
```

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employee_id	name	salary	bonus	total compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

5 The projects table

14 Show total and average budget per department. Only include departments with average budget above 70 000.

```
SELECT  department
        SUM (budget) AS total_budget
        AVG (budget) AS avg_budget
FROM    projects
GROUP BY department
HAVING AVG (budget) > 70 000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	

15. List projects with budgets between 50 000 and 120 000, excluding the Marketing department

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```
SELECT *
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
      AND department != 'Marketing';
```

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000